**Part A Mini Project**
**Multiplication of matrix**
**Code**

```
#include <bits/stdc++.h>
#include <pthread.h>
using namespace std;

#define MAX 4

int A[MAX][MAX], B[MAX][MAX], C[MAX][MAX];
int row = 0;
pthread_mutex_t lock1 = PTHREAD_MUTEX_INITIALIZER;

void* multiply(void* arg) {
    pthread_mutex_lock(&lock1);
    int i = row++;
    pthread_mutex_unlock(&lock1);

    for (int j = 0; j < MAX; j++)
        for (int k = 0; k < MAX; k++)
            C[i][j] += A[i][k] * B[k][j];

    return NULL;
}

int main() {
    srand(1);

    for (int i = 0; i < MAX; i++)
        for (int j = 0; j < MAX; j++) {
            A[i][j] = rand() % 10;
            B[i][j] = rand() % 10;
        }

    cout << "Matrix A\n";
    for (auto &r : A) {
        for (int x : r) cout << x << " ";
        cout << endl;
    }

    cout << "\nMatrix B\n";
    for (auto &r : B) {
        for (int x : r) cout << x << " ";
```

```
        cout << endl;
    }

    pthread_t t[MAX];

    for (int i = 0; i < MAX; i++)
        pthread_create(&t[i], NULL, multiply, NULL);

    for (int i = 0; i < MAX; i++)
        pthread_join(t[i], NULL);

    cout << "\nMultiplication of A and B\n";
    for (auto &r : C) {
        for (int x : r) cout << x << " ";
        cout << endl;
    }
    pthread_mutex_destroy(&lock1);
    return 0;
}
```

**Output**

```
Matrix A
3 7 3 6
9 2 0 3
0 2 1 7
2 2 7 9

Matrix B
6 5 5 2
1 7 9 6
6 6 8 9
0 3 5 2

Multiplication of A and B
43 100 132 87
56 68 78 36
8 41 61 35
56 93 129 97
```